

White Paper on AI Governance

Leadership insights and the Voluntary AI
Safety Standard in practice.



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About the Governance Institute of Australia

A national membership association, Governance Institute of Australia advocates for a community of governance and risk management professionals, equipping over 8,000 members with the tools to drive better governance within their organisation. We tailor our resources for members in the listed, unlisted, not-for-profit and public sectors, and ensure our members' voice is heard loudly. As the only Australian provider of chartered governance accreditation, we offer a range of short courses, certificates and postgraduate study to help further the knowledge and education of the fast-growing governance and risk management profession. We run a strong program of thought leadership, research projects and news publications and draw upon our membership of the Chartered Governance Institute to monitor emerging global trends and challenges to ensure our members are prepared. Our members know that governance is at the core of every organisation – and in these tumultuous times, that good governance is more important than ever before.



About the National Artificial Intelligence Centre

The National AI Centre (NAIC) was established in 2021 to support and accelerate Australia's AI industry. It aims to help Australia become a global leader in developing and adopting safe and responsible artificial intelligence. NAIC is doing this by: supporting AI adoption for small and medium businesses by addressing barriers and challenges, growing an Australian AI industry, convening the AI ecosystem, uplifting safe and responsible AI practice.

Foreword

In today's rapidly evolving technological landscape, artificial intelligence (AI) stands at the forefront of innovation, transforming industries and redefining the way we live and work. I am thrilled to present this joint thought leadership report in collaboration with the National AI Centre, which delves into the profound impact of AI and its potential to shape our future.

This report is a culmination of three specialist roundtables with key insights from industry experts, and real-world case studies that highlight the transformative power of AI. It explores the latest trends, challenges, and opportunities in AI, providing a comprehensive overview of how businesses can harness this technology to drive growth, enhance efficiency and create value.

It also examines the critical role of governance in the ethical and responsible deployment of AI. Good governance is essential to ensure that AI technologies are developed and used in ways that are transparent, fair, and aligned with societal values. Our partnership with NAIC underscores our commitment to not only advancing AI but doing so in a manner that upholds the highest standards of integrity and accountability.

We are proud to share our knowledge and expertise through this report, aiming to inspire and guide organisations on their AI journey.

I would like to extend my gratitude to the talented team of researchers, analysts, and contributors who have made this report possible. Their hard work and dedication have resulted in a valuable resource that we hope will serve as a catalyst for meaningful discussions and strategic initiatives in the AI domain.

I would also like to sincerely thank our sponsored partners, Clayton Utz, Diligent and PKF for their support in bringing this report to life and for their continued involvement with the work of the Governance Institute of Australia.

As you explore the insights and findings within this report, I encourage you to think boldly about the future of AI and its potential to drive positive change. Together, we can harness the power of AI to create a smarter, more connected, and more prosperous world.

Thank you for your interest in our AI thought leadership report. We look forward to continuing this journey of innovation and discovery with you.



Megan Motto FGIA FCG FAICD

Chief Executive of Governance Institute of Australia Ltd and of the Australian Division of The Chartered Governance Institute

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CLAYTON UTZ

Proudly Australian and globally connected, Clayton Utz gives confident, innovative and incisive advice beyond clear matters of law. We are one of Australia's oldest and largest law firms, with offices in six cities, nearly 200 partners, and provide a full range of legal services for large private sector and government clients. At the forefront of legal innovation we blend technology with legal acumen to provide inventive commercial solutions.

"The extraordinary advancements in AI technology in recent years have triggered a global conversation about how AI is governed and regulated. Governance and regulation play a critical role in the safe and responsible development and use of AI technologies - to build consumer trust and encourage adoption. Applying appropriate governance, compliance and risk management to policies, projects and procurement should be an essential component of any successful adoption of AI."

Diligent



Diligent is the leading GRC SaaS company, empowering more than 1 million users and 700,000 board members and leaders to make better decisions, faster. The Diligent One Platform helps organisations connect their entire GRC practice - including governance, risk, compliance, audit and ESG - to bring clarity to complex risk, stay ahead of regulatory changes and deliver impactful insights, in one consolidated view.

"Continuous risk management and stakeholder engagement are key recommendations that resonate deeply with our practices at Diligent. Ensuring that AI deployments are in line with organisational goals and ethical considerations is essential for sustainable and responsible governance."

PKF



PKF is part of a global network, where dynamic business advisors can belong, grow, and thrive. In Australia, with more than 100 partners and 800 talented people, they deliver advisory, audit and tax solutions to create powerful opportunities for their clients, their people and their communities.

A can do attitude & strong project management skills lie at the heart of any successful initiative & this would equally apply to any move to use AI. Poor disciplines more broadly will lead to poor outcomes which could be accelerated through technology such as AI. Similarly, a strong data governance framework should enhance & secure productivity & innovation gains from responsible use of AI—not forgetting the overriding responsibility of quality control, quality management & earning the right to rely on its outcomes.

List of Roundtable participants:

AI ethics and governance

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Executive Summary

In today's rapidly evolving technological landscape, Australian business leaders face a unique challenge and opportunity: the adoption of Artificial Intelligence (AI). Despite AI's transformative potential, confidence and adoption rates among Australian businesses lag behind those in comparable countries. It is this very issue that we explore in this report by surfacing leadership insights gathered from roundtables, and aligning best practice from the recently released Voluntary AI Safety Standard.

Australian businesses are not fully leveraging AI, leading to missed opportunities for innovation and efficiency. Additionally, AI presents significant ethical dilemmas that require thoughtful leadership and robust frameworks. This document offers strategic recommendations and practical guidance to help business leaders navigate AI adoption. We present a tailored approach that considers the specific needs and challenges of Australian businesses, from SMEs to larger firms.

Data-Driven Insights: We provide up-to-date statistics and comparative analysis to highlight the current state of AI adoption in Australia.

Ethical Considerations: The report delves into the ethical implications of AI, offering approaches to address these challenges responsibly.

Actionable Recommendations: Clear, actionable steps from the latest frameworks are outlined to help business leaders implement AI effectively and ethically.

This report is designed for organisational and business leaders across various sectors, including SMEs and larger enterprises. It aims to equip them with the knowledge and tools needed to lead their organisations through the AI transformation.

This report complements additional guidance published by the Australian Government, in particular, the Voluntary AI Safety Standard.

Understanding and adopting AI is crucial for staying competitive in today's market. Through this report, business leaders will gain valuable insights and practical strategies to harness the power of AI safely and responsibly, ensuring their organisations remain at the forefront of innovation.

Recommendations in this report

Throughout this report we have extracted two types of recommendations. The best practice guardrail from the Voluntary AI Safety Standard that was recently published by the National AI Centre, and the Expert Tip which came from the leadership insights during the roundtable discussions as well as from our partners.

Voluntary AI Safety Standard



Guardrail 1:

Establish, implement and publish an accountability process including governance, internal capability and a strategy for regulatory compliance.



Expert tip:

Consider how AI can help monitor and achieve ESG goals, and how ESG principles can act as a first step to inform the design of AI governance frameworks.

Voluntary AI Safety Standard¹

Summary of 10 Guardrails

1. [Establish, implement and publish an accountability process including governance, internal capability and a strategy for regulatory compliance.](#)

Guardrail one creates the foundation for your organisation's use of AI. Set up the required accountability processes to guide your organisation's safe and responsible use of AI, including:

- an overall owner for AI use
- an AI strategy
- any training your organisation will need.

2. [Establish and implement a risk management process to identify and mitigate risks.](#)

Set up a risk management process that assesses the AI impact and risk based on how you use the AI system. Begin with the full range of potential harms with information from a stakeholder impact assessment (guardrail 10). You must complete risk assessments on an ongoing basis to ensure the risk mitigations are effective.

3. [Protect AI systems, and implement data governance measures to manage data quality and provenance.](#)

You must have appropriate data governance, privacy and cybersecurity measures in place to appropriately protect AI systems. These will differ depending on use case and risk profile, but organisations must account for the unique characteristics of AI systems such as:

- data quality
- data provenance
- cyber vulnerabilities.

4. [Test AI models and systems to evaluate model performance and monitor the system once deployed.](#)

Thoroughly test AI systems and AI models before deployment, and then monitor for potential behaviour changes or unintended consequences. You should perform these tests according to your clearly defined acceptance criteria that consider your risk and impact assessment.

5. [Enable human control or intervention in an AI system to achieve meaningful human oversight across the life cycle.](#)

It is critical to enable human control or intervention mechanisms as needed across the AI system lifecycle. AI systems are generally made up of multiple components supplied by different parties in the supply chain. Meaningful human oversight will let you intervene if you need to and reduce the potential for unintended consequences and harms.

¹ National AI Centre, [Voluntary AI Safety Standard](#)

6. [Inform end-users regarding AI-enabled decisions, interactions with AI and AI-generated content.](#)

Create trust with users. Give people, society and other organisations confidence that you are using AI safely and responsibly. Disclose when you use AI, its role and when you are generating content using AI. Disclosure can occur in many ways. It is up to the organisation to identify the most appropriate mechanism based on the use case, stakeholders and technology used.

7. [Establish processes for people impacted by AI systems to challenge use or outcomes.](#)

Organisations must provide processes for users, organisations, people and society impacted by AI systems to challenge how they are using AI and contest decisions, outcomes or interactions that involve AI.

8. [Be transparent with other organisations across the AI supply chain about data, models and systems to help them effectively address risks.](#)

Organisations must provide information to other organisations across the AI supply chain so they can:

- understand the components used including data, models and systems
- understand how it was built
- understand and manage the risk of the use of the AI system.

9. [Keep and maintain records to allow third parties to assess compliance with guardrails.](#)

Organisations must maintain records to show that they have adopted and are complying with the guardrails. This includes maintaining an AI inventory and consistent AI system documentation.

10. [Engage your stakeholders and evaluate their needs and circumstances, with a focus on safety, diversity, inclusion and fairness.](#)

It is critical for organisations to identify and engage with stakeholders over the life of the AI system. This helps organisations to identify potential harms and understand if there are any potential or real unintended consequences from the use of AI. Deployers must identify potential bias, minimise negative effects of unwanted bias, ensure accessibility and remove ethical prejudices from the AI solution or component.

Introduction



Artificial intelligence (AI) has emerged as a transformative force across various sectors. It is driving *innovation and efficiency* at an unprecedented scale. However, the rapid advancement (such as General Purpose AI) and integration of AI technologies into our daily lives present significant challenges and opportunities that demand careful examination.

This is also underscored by key policy discussions, such as the US Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence, the Bletchley Declaration, and the European Union (EU) AI Act – the first of its kind – reflecting a global consensus on this critical issue.

Understanding AI in the Australian context

Approaching AI safety and responsible AI (RAI) necessitates *joint efforts* from multidisciplinary communities such as AI, software engineering (SE) and governance. To better understand Australia's stance on RAI, we hosted 3 roundtable discussions with panelists from different fields.

These session topics covered:

- AI ethics and governance
- AI risk and opportunity
- AI as an enabler.

Each session explored crucial aspects of AI development and deployment, providing a holistic understanding of the implications, and the potential and responsibilities associated with AI technologies. The findings and recommendations from those discussions are presented across the three core topics in this white paper.



Roundtable themes

1. AI ethics and governance

Explores the fundamental principles and frameworks needed to ensure that AI systems are developed and deployed ethically and responsibly.

It addresses the moral, legal and societal implications of AI, emphasising the importance of AI ethics principles, their implementation and operationalisation, and governance for AI.

2. AI risk and opportunity

Examines the dual facets of AI. It highlights the potential risks, including biases, security concerns and unintended consequences, while also exploring the immense opportunities AI presents for innovation, problem-solving and enhancing human capabilities.

3. AI as an enabler

Investigates how AI can be a powerful tool for enabling progress across various domains. It focuses on the transformative impact of AI on industries, the economy and society at large, showcasing the role of AI in driving sustainable development and addressing complex global challenges.

1. AI ethics and governance

The rapid advancement of AI underscores the necessity for robust ethical frameworks and governance mechanisms. AI ethics and governance have emerged as essential areas of study and practice, aiming to ensure that AI technologies are developed and deployed in ways that are responsible and safe.

There are a wide range of considerations for [AI ethics](#). For instance, Australia's AI Ethics Principles² call on organisations to:

- avoid bias and discrimination
- understand and adopt cultural and linguistic diversity and gender equality
- drive confidence through reliable, safe and accurate decision-making.

Ultimately, they present the fundamental 'do no harm principle'.

AI ethics involves a meticulous examination of the moral implications of AI decisions, the responsibilities of AI developers and users, and the societal consequences of AI deployment. The ethical dimension of AI requires interdisciplinary approaches, drawing from philosophy, law, sociology and computer science, to address the multifaceted challenges posed by these technologies.

[AI governance](#) refers to the frameworks, policies and regulations that oversee the development and implementation of AI systems. Effective AI governance aims to establish standards and guidelines that promote ethical AI practices, protect individual rights and mitigate risks associated with AI. It involves the collaboration of various stakeholders, including governments, international organisations, industry leaders and society, to formulate and enforce policies that ensure AI systems are used responsibly and for the benefit of all.

The [intersection](#) of AI ethics and governance is crucial in addressing the profound impacts of AI on society.

As AI systems become increasingly autonomous and complex, the potential for unintended consequences and ethical dilemmas grows. That is why it is imperative to establish comprehensive governance structures to address current ethical concerns while being adaptable to future technological advancements.

² Australian Government Department of Industry, Science and Resources, n.d. [Australia's AI Ethics Principles](#).



Deciding to use AI

AI holds *transformative potential* across various sectors, enabling advanced operational efficiencies that can redefine industry standards, yet the motivations for integrating AI vary significantly among organisations. The justification for AI deployment must be scrupulously crafted to align with specific organisational contexts, cultures and strategic goals. It is crucial for each organisation to precisely identify its unique drivers for AI adoption, which may range from enhancing operational productivity to securing tangible competitive advantages in the marketplace.

The strategic imperative of AI is further highlighted by its necessity for maintaining competitiveness in a rapidly evolving technological landscape. A reluctance or delay in embracing cutting-edge technologies like AI can result in significant missed opportunities and a risk of falling behind other organisations. This urgency is fuelled by both external market dynamics and internal demands for continual innovation. Notably, younger employees, often more conversant with emerging technologies, play a pivotal role in advocating for AI integration. Their enthusiasm for technological assimilation compels organisations to embed AI strategies within their core initiatives for sustained relevance and innovation.

Aligning AI deployment with organisational purposes is essential for strategic success.

Organisations must ensure that AI adoption not only supports their strategic goals but also enhances operational efficiency. This strategic alignment is crucial as it directly influences an organisation's ability to leverage AI to its full potential, optimising processes and catalysing innovation in service delivery and product development. For instance, companies in the financial services sector tend to adopt AI more cautiously, updating data and information security policies to harness AI capabilities while ensuring compliance with evolving regulatory standards.

The drive to adopt AI transcends merely staying current with technology trends; it is fundamentally integrated into strategic imperatives that dictate competitive advantage and operational excellence. Yet successfully adopting and integrating new AI technologies involves navigating a complex landscape of challenges, including:

- data privacy concerns
- the need for substantial investments in skills and infrastructure
- managing ethical considerations.

By embedding 'the value-add' of AI into strategic frameworks or business models, organisations can address these challenges while leveraging the full potential of AI to achieve transformative operational capabilities. This strategic integration ensures that organisations can assess the benefits and associated costs and risks, meaning they can meet the immediate demands for innovation but also position

Voluntary AI Safety Standard



Guardrail 1:

Establish, implement and publish an accountability process including governance, internal capability and a strategy for regulatory compliance.

themselves at the forefront of their respective industries. Effective harnessing of AI capabilities, as detailed in Topic 2, offers both challenges and opportunities, necessitating robust RAI and AI governance mechanisms to foster an environment conducive to innovative growth and strategic enhancement.

RAI and AI ethical principles for decision-making

As AI becomes increasingly integral to organisational operations and strategies, the importance of RAI and ethical principles escalates. Various sets of *principles* have been developed by different organisations and government agencies worldwide, such as Australia's AI Ethics Principles³, OECD AI Principles⁴ and Hiroshima AI Process⁵. These principles have a similar fundamental focus on areas of fairness, transparency, accountability and explainability.

These principles reflect the wide range of ethical dilemmas and operational challenges that AI can introduce. Serving as essential high-level guidelines, they ensure that AI systems operate responsibly and safely. However, due to their *abstract nature*, these principles often require further operationalisation to be effectively implemented within specific contexts. This process of transforming high-level ethical guidelines into actionable operational practices is crucial as it ensures that AI systems adhere to established ethical norms and are tailored to meet the nuanced requirements of diverse organisational environments and societal expectations.⁶ Some of the shared fundamental principles are explored below.

ESG and human-centered design

Integrating Environmental, Social and Governance (ESG) criteria into AI development is essential for a comprehensive evaluation of a technology's broader impacts. ESG is a good starting block for considering how to effectively manage AI across an organisation – ensuring AI initiatives promote sustainability, ethical governance and social responsibility. AI can also assist organisations to achieve their ESG goals and ambitions. AI technologies can be leveraged to provide useful insights into ESG metrics, such as identifying and mitigating unethical labour practices and unsustainable material sourcing across the supply chain.

In addition to ESG, prioritising *human-centered design* within this framework ensures that AI systems are accessible, intuitive and designed with the end-user's welfare in mind, thereby enhancing user experience and adherence to ethical norms.⁷

3 Australian Government Department of Industry, Science and Resources, n.d. [Australia's AI Ethics Principles](#).

4 OECD, 2019 (updated May 2024), [OECD AI Principles](#).

5 Google, n.d. [AI Principles](#).

6 Xia B, Lu Q, Perera H, Zhu L, Xing Z, Liu Y and Whittle J, May 2023, 'Towards Concrete and Connected AI Risk Assessment (C²AIRA): A Systematic Mapping Study', *2023 IEEE/ACM 2nd International Conference on AI Engineering-Software Engineering for AI (CAIN)*:104–116, IEEE.

7 Alphinity Investment Management and CSIRO, 2024, [The intersection of Responsible AI and ESG: A Framework for Investors](#).

ESG becomes a good framework to consider AI risks because they know how to report it. Many risks already captured in ESG are relevant for AI, such as emission consequences, social and governance risks [...] We need to look at additional AI risks that the board needs to be careful about. Some risks are wholly new and not covered in traditional ESG, especially in the social and governance aspects.

– Liming Zhu

Fairness, transparency and explainability

Achieving fairness in AI necessitates that systems are thoughtfully designed and undergo rigorous, regular audits to identify and mitigate biases, ensuring equitable outcomes for all users. This commitment to **fairness** must be embedded from the initial design phase through to deployment and beyond, with continuous assessments to adapt to evolving data and contexts.

Transparency complements this commitment, requiring that the operations and decision-making processes of AI systems are clearly articulated and accessible to users. This clarity is vital, as it not only builds **trust and confidence** among users and stakeholders but also ensures that AI actions are comprehensible and defensible. A striking illustration of the importance of these principles is a study which identified gender biases in 44% of 133 AI systems analysed.⁸ This statistic highlights a significant concern and underscores the critical need for robust mechanisms to ensure fairness and transparency in AI. By prioritising these principles, organisations can provide stakeholders with insights into the inner workings of AI technologies, demystifying AI processes and reinforcing the ethical integrity of their applications.

Bias can creep into an AI selection tool in a number of ways. For example, there can be:

- historical bias
- sampling bias
- measurement bias
- evaluation bias
- aggregation bias
- deployment bias⁹



Expert tip:

Consider how AI can help monitor and achieve ESG goals, and how ESG principles can act as a first step to inform the design of AI governance frameworks.

These biases could cause the AI to produce output which is harmful or violate anti-discrimination laws. This is particularly problematic when AI is used to make decisions or given agency to act in an autonomous way. These risks require human oversight and accountability to prevent

8 Smith, G and Rustagi, I 2021 "[When good algorithms go sexist, why and how to advance AI gender equity](#)."

9 Sheard N, 2022, '[Employment Discrimination by Algorithm: Can Anyone be Held Accountable?](#)', *UNSW Law Journal*, Volume 45(2):617–648.

and mitigate them, for example, by reviewing and adjusting the outcome to account for any biases. There have been some instances of this problem in the context of recruitment. For example, a claim brought against Workday alleged that an AI screening tool made available by Workday discriminated against applicants on the basis of race, disability, and age – each of which are protected attributes under discrimination laws.

Privacy and cyber security

In the digital age, where data breaches are increasingly frequent, establishing robust privacy and security protocols for AI systems is crucial. These systems must implement stringent measures to *safeguard* personal and sensitive information against unauthorised access, aligning with the Privacy Act and global data protection regulations such as the General Data Protection Regulation (GDPR). In the context of generative AI, it is vital to address copyright and licensing issues rigorously. AI systems often create new content by learning from vast datasets that are often crawled online¹⁰. Similarly, organisations using their enterprise data with generative AI must ensure that the use of such data complies with privacy, confidentiality and copyright laws and licensing agreements. This commitment not only secures stakeholder trust but also upholds the integrity of data and content created or processed by AI, emphasising an organisation's dedication to ethical data usage and intellectual property rights.

The use of third-party material or personal information to train, prompt or ground AI could violate Australian copyright or privacy laws. In particular:

- The process of training AI often involves a reproduction of copyright material in the training data. Using a model may generate reproductions of the training data or reproduce other material to ground or prompt the model. These reproductions may infringe copyright if there is no appropriate licence or statutory exception.
- To the extent training data or prompting data includes personal information, privacy laws impose a restriction on how that data may be collected, used and disclosed.

In Australia, copyright laws are not currently broad enough (with some limited exceptions) to allow for the use of data to train AI models without an appropriate *licence*. Globally, copyright issues have resulted in a wave of litigation against AI companies that is currently making its way through the courts. The AI industry is also dealing with this issue in a number of ways, including by negotiating licences with content creators and by providing guarantees to consumers that their products are not infringing.



Expert tip:

Implement a continuous improvement feedback model to identify and mitigate biases and drive fairer outcomes for all users interacting with it.

¹⁰ Zhang D, Xia B, Liu Y, Xu X, Hoang T, Xing Z, Staple M, Lu Qinghua and Zhu L, 2023, 'Navigating privacy and copyright challenges across the data lifecycle of generative ai', arXiv preprint arXiv:2311.18252.

Voluntary AI Safety Standard



Guard rail 3:

Protect AI systems, and implement data governance measures to manage data quality and provenance.

A notable example of training AI in violating privacy laws is the case of Clearview AI – an AI company that used images scraped from the internet to create biometric information stored in its facial recognition database, a practice which the Office of the Australian Information Commissioner declared to have breached the *Australian Privacy Act 1988*.

The use of AI also creates new security challenges for organisations and individuals. This is due in part to the need for the AI to access – and therefore for the organisation to retain – large quantities of data. The retention of this data increases the risk and potential consequences of a security breach. So, organisations should ensure appropriate controls are in place to prevent and mitigate the risk.

Technological advancements in AI may also increase the opportunity for malicious activity. For example, the potential to create higher quality deepfakes – being hyper realistic but false depictions of a person or thing – which may aid cybercrime, including identity theft and phishing, or spread misinformation, among other things.

Reliability and safety

Ensuring the reliability and safety of AI is a fundamental concern across all branches of AI. AI models and/or systems must undergo rigorous testing and validation to perform reliably under diverse conditions without introducing unforeseen risks. Continuous monitoring and thorough safety evaluations are essential, not only to adhere to regulations such as those under consideration by the Australia Government¹¹ or those in the *European Union Artificial Intelligence Act* (EU AI Act) but also to maintain trust and operational integrity.

As we delve into the realm of generative AI, these concerns become even more pronounced due to its versatile capabilities. Generative AI, which can produce text, images and other content, requires additional layers of scrutiny. The capability evaluations for these systems are particularly critical – to assess the ability of generative AI and other branches of AI to handle tasks safely and effectively. Importantly, these evaluations cover both *intended capabilities* envisioned by developers and *unintended capabilities* that could emerge as byproducts of complex interactions within the AI model. This comprehensive approach ensures that the outputs are not only high quality but also ethically sound and free from harmful content. Given their potential impact, it is imperative that these evaluations are comprehensive (for example, both model- and system-level, with considerations across the AI supply chain¹²) and adapt to the rapid advancements in generative AI.

In this context, continuous adaptation to emerging threats and the ability to respond to anomalies swiftly are paramount. This ensures that generative AI systems not only comply with stringent safety standards but also uphold the ethical standards necessary for their widespread adoption in sensitive and impactful domains. This is particularly necessary in high-risk settings such as medical diagnostics and road safety.

¹¹ Introducing mandatory guardrails for AI in high-risk settings: proposals paper - Consult hub (industry.gov.au)

¹² Xia B, Lu Q, Zhu L and Xing Z, 2024, 'Towards AI safety: A taxonomy for ai system evaluation', arXiv preprint arXiv:2404.05388.

Voluntary AI Safety Standard



Guardrail 4:

Test AI models and systems to evaluate model performance and monitor the system once deployed.

Ensuring the reliability and safety of AI systems is important in complying with many legal obligations. For example, inaccurate information could lead to claims of negligence, breach of contractual warranties, breach of Australian consumer laws (including laws preventing misleading or deceptive conduct) or defamation. Many sector-specific laws may also apply (such as regulation of Software as a Medical Device).

Contestability and accountability

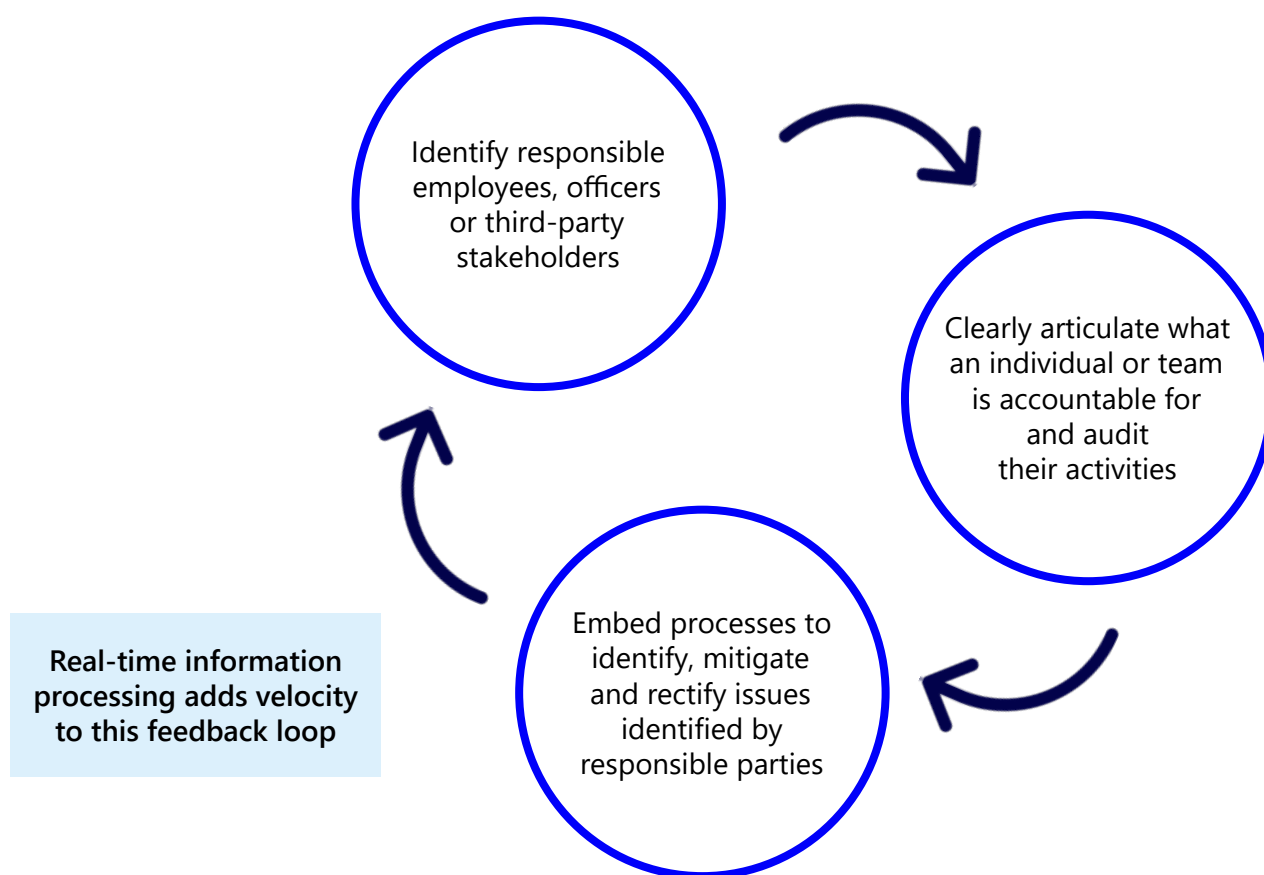
AI systems must enable affected parties to contest decisions effectively, providing a robust process for airing grievances and implementing corrections. This capacity for contestability ensures overall safety in AI decision-making processes. Accountability complements this by requiring clear organisational governance structures, including clear identification of roles and responsibilities, and protocols to manage and rectify any adverse outcomes or errors. This principle is essential for upholding ethical standards and enhancing organisational credibility.

To effectively operationalise accountability in AI systems, it is critical to structure it around 3 core pillars:

- **Responsibility** – defines ‘who is accountable and to whom’, establishing clear roles and responsibilities among developers, users and other stakeholders. This foundational aspect sets the stage for measurable and enforceable accountability.
- **Auditability** – focuses on ‘what one is accountable for’. This ensures that all actions and decisions made by AI systems are traceable and auditable, which is essential for maintaining transparency and facilitating the evaluation of AI systems against agreed standards and regulations.
- **Redressability** – addresses ‘how entities are accountable and can rectify issues’. It includes establishing mechanisms for correcting any mistakes or misjudgements made by AI systems and providing effective remedies for those adversely affected.¹³

¹³ Xia B, Lu Q, Zhu L, Lee SU, Liu Y and Xing Z, 2023, ‘Towards a Responsible AI Metrics Catalogue: A Collection of Metrics for AI Accountability’, arXiv preprint arXiv:2311.13158.

Figure 1: Implementing contestable accountability processes in AI systems: A 3-pillar approach



Expert tip:

Adopt the 3-pillar approach to implement contestability and accountability in AI systems by identifying responsible employees, agents or entities, articulating expectations of their activities and auditing their actions, and embedding processes to assist in the identification, mitigation and rectification of issues.

The increased use of AI in decision-making introduces a range of issues company directors and other responsible persons must consider when discharging their existing duties under the Australian *Corporations Act 2001*, notably, the duty under s180 to exercise reasonable care and diligence. Directors may be accountable for an outcome, even if the decision-making process has been automated.

Similar issues exist in government. Administrative decision-makers must afford procedural fairness, must only act within the scope of their decision-making power and must only consider relevant considerations – any use of AI must not detract from those principles. Further, in some cases, decisions are legally required to be made by a person. In those cases, a decision made by AI will be invalid.

AI governance

Effective governance of AI is essential to ensure that AI technologies are developed, deployed and used responsibly, adhering to safety and ethical norms, and complying with regulatory standards. AI governance involves comprehensive practices and policies that guide the lifecycle of AI systems, aimed at mitigating risks and maximising societal benefits.

Approach to AI governance

Strategic oversight and ethical alignment: AI governance starts with strategic oversight, involving the setting of clear goals and ethical guidelines for AI use within organisations. This includes aligning AI strategies with the organisation's overarching goals and a firm commitment to ethical practices at all levels. Leaders must champion a culture of ethical integrity and accountability, ensuring AI solutions are responsibly developed and used. AI governance should form part of an AI Risk Management Framework.¹⁴

If artificial intelligence is to be used safely and responsibly within an organisation, we know that it is vital for leaders to be actively involved in shaping policies and procedures.

– Stela Solar

Regulatory compliance and ethical standards: Compliance with local and international regulations is essential. AI governance frameworks must adapt to the evolving legal landscape, incorporating standards that address new ethical challenges such as privacy concerns, bias mitigation and transparency. Effective governance requires creating regulatory settings that support the safe and responsible adoption of AI while fostering innovation.

Risk assessment and management: Central to AI governance is the rigorous management of potential risks, covering both technical vulnerabilities and ethical dilemmas. AI systems must be evaluated for their impact on fairness, privacy and security. Effective risk management strategies involve identifying and mitigating immediate risks and anticipating future challenges as technology and societal norms evolve.

If you manage a risk better than your competitors, then it's an opportunity.

– George Gorman, Zip Co.



Expert tip:

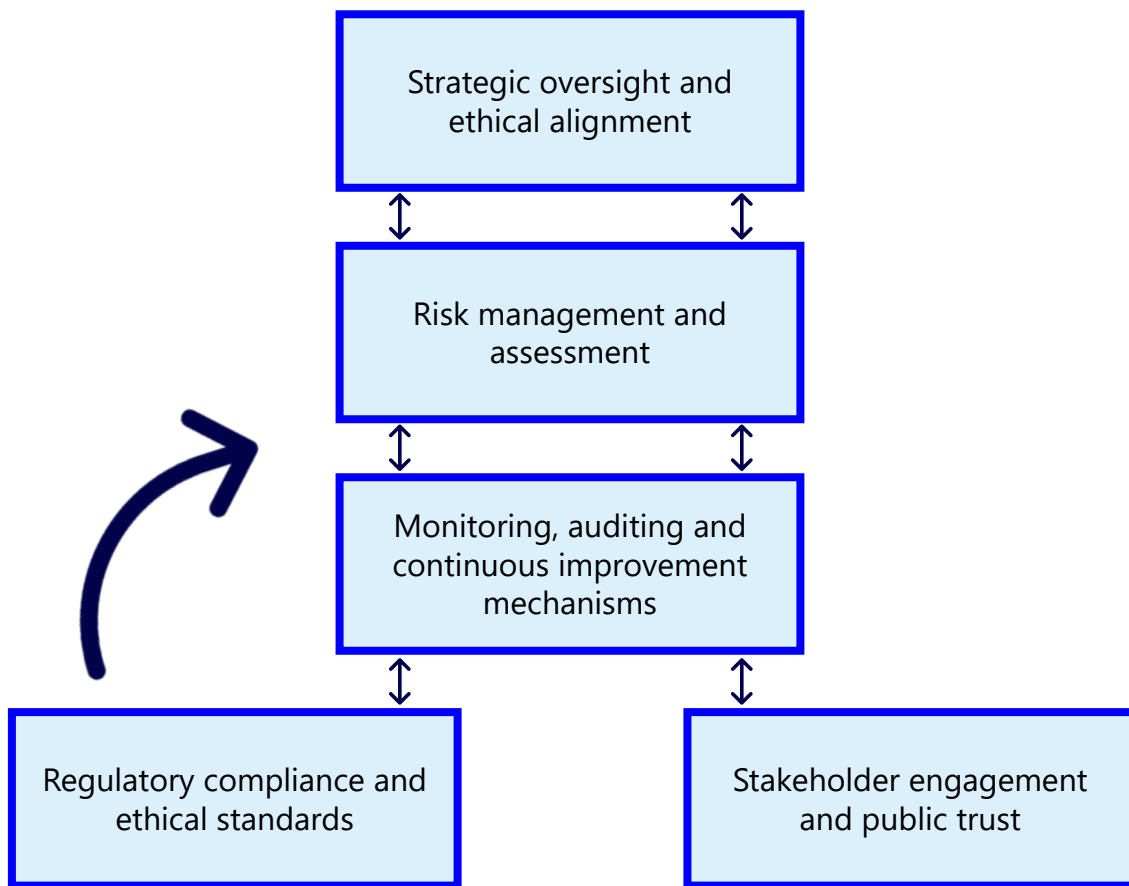
Report issues to senior officers and directors when known issues may have financial or reputational harm to the business, particularly as it relates to s 180 Duty of Care and Diligence and ASX Listing Rule 3.1 regarding Continuous Disclosure.

Stakeholder engagement and public trust: Continuous engagement with a broad range of stakeholders, including technologists, policymakers, affected communities and the public is crucial. This engagement ensures that diverse perspectives and values inform AI development and deployment. Building public trust through transparent practices and open communication is critical for the broad acceptance and successful integration of AI systems.

Monitoring, auditing and continuous improvement: Ongoing monitoring and regular auditing of AI systems are essential to ensure they operate as intended and adhere to ethical standards over time. This process of continuous assessment and adaptation keeps AI governance frameworks responsive to rapid technological advancements and changing societal expectations.

¹⁴ NIST, 2023, AI Risk Management Framework.

Figure 2: Inclusive AI governance framework



Challenges in AI governance

AI governance faces several significant challenges that can complicate the effective management and oversight of these technologies:

- **Complexity of AI systems:** The inherent complexity of AI algorithms and their decision-making processes can make transparency and accountability difficult. This complexity often requires specialised knowledge, posing a barrier for stakeholders attempting to understand or oversee AI systems. 'Black box' system processes affect trust and confidence across the AI supply chain and create hesitancy in risk-taking by senior officers and directors, potentially stifling business innovation.
- **Rapid technological advancement:** The fast pace of AI development can outstrip current governance frameworks and regulatory guidelines, making it difficult to keep up with new technologies and their potential impacts. Real-time information and continuous feedback loops can assist where technological capabilities are being stretched into new business domains.



Expert tip:

Implement an inclusive AI governance framework with continuous feedback loops that address key regulatory compliance and ethical standards frameworks.

- **Global disparities in regulations:** Differences in AI regulations across countries can lead to governance gaps, especially for multinational corporations operating in multiple jurisdictions. Harmonising these differences remains a formidable challenge. However, fundamental principles and concepts of safe and responsible AI development and deployment are gaining traction.
- **Ethical ambiguities:** Ethical standards in AI are continually evolving, and there can be significant disagreements on what constitutes ethical use, particularly in areas like facial recognition and predictive policing. Social norms will continue to evolve and change over time. Organisations should be aware of their consumer and supplier risk appetite and ethical parameters.
- **Resource allocation:** Adequate resources, including funding, expertise and tools, are essential for effective AI governance but are often lacking, especially in smaller organisations or in sectors with rapid AI integration. Capability uplift across the organisation in otherwise non-technical roles can be leveraged to address the resource gap.

By addressing these challenges head-on, organisations can enhance their AI governance frameworks, ensuring that AI technologies are leveraged responsibly and ethically across all applications.



2. AI risks and opportunities

Voluntary AI Safety Standard



Guardrail 6:

Inform end-users regarding AI-enabled decisions, interactions with AI and AI-generated content.

The integration of AI into various sectors presents a unique blend of risks and opportunities that organisations must skilfully navigate. This section explores the dual nature of AI's impact, focusing on the *practical implications* for business and society, offering a more pragmatic exploration of how AI reshapes industries and introduces new challenges.

AI technologies introduce risks that require vigilant management, including the risk of disseminating or relying on inaccurate, unreliable or biased/discriminatory content, intellectual property infringement risks and data privacy risks where personal information might be compromised, if not designed and deployed with clear ethical guidelines, generative AI can have unintended consequences and potentially cause real harm.¹⁵

The opportunities presented by AI are *transformative*. AI facilitates the automation of complex tasks, yields insightful analytics from large datasets, and enables unprecedented levels of personalisation in customer service and product offerings. For instance, generative AI revolutionises creative processes and content generation, leading to significant operational efficiencies, cost reductions and enhanced market competitiveness.

By strategically leveraging AI, organisations can improve their internal processes, redefine customer interactions and expand into new markets, driving substantial business growth.

Governments understand the need to balance innovation and risk-taking with precautionary principles that factor risks associated with AI. International governments have taken different approaches in managing risks and opportunities, with some jurisdictions taking a more precautionary approach to AI risk management, creating clear limits on high-risk activities such as those outlined in the EU AI Act. The US Government also recognised the potential for significant harm of unregulated AI by creating the *Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence*. The Australian government has published a Mandatory Guardrails Proposals Paper open for consultation¹⁶. Governments are aware of the tremendous benefits to society and the economy that AI can bring but remain cognisant to the potential harms of unabated AI development.

¹⁵ [Managing the Risks of Generative AI \(hbr.org\)](https://hbr.org/2023/01/managing-the-risks-of-generative-ai)

¹⁶ [Australian Government, Mandatory Guardrails Proposal Paper](https://www.australian.gov.au/ai/guardrails)



Guardrail 2:

Establish and implement a risk management process to identify and mitigate risks.

Australia is also following this precautionary approach. AI regulation has been on the Government's agenda since the release of its discussion paper 'Safe and responsible AI in Australia' in June 2023 which was then followed by an interim response paper in January 2024 signalling that mandatory regulation would follow. At the time of writing this paper in September 2024, the Australian Government has released a [consultation paper](#) proposing the introduction of regulations requiring 10 mandatory guardrails focusing on testing, transparency and accountability to be adopted by Australian organisations in developing or deploying high risk AI systems.

AI risks

The development, deployment and use of AI technologies presents a complex interplay between risks and opportunities, necessitating vigilant management and strategic governance. These risks are intertwined with opportunities, highlighting the need for a balanced approach to harness AI's full potential while meeting ethical standards.

Integrated risk and opportunity management

Managing AI involves a nuanced approach that starts with testing initial value propositions iteratively, then adjusting strategies based on comprehensive risk and impact assessments. This *incremental and iterative* method helps identify and mitigate risks early, ensuring that AI implementations are deliberate and considerate. For example, the deployment of AI in loan approval processes can be tested in controlled environments to identify unintended biases against certain demographics before full-scale implementation.

Scaling AI risk management

As AI technology becomes more accessible, the challenge of managing its risks also scales. The expansion of AI capabilities increases the scope of potential impact, meaning that errors or missteps in AI applications can have consequences that extend beyond affecting just small groups of individuals. These errors can have wide-reaching effects, potentially impacting entire populations or causing systemic failures across entire networks or sectors. This underscores the need for *robust standards* and training for the safe development, deployment and use of AI – and these must be comprehensible to non-specialists. As AI becomes more accessible to individuals and employees, the risk of improper use and integration of AI into organisational tasks and activities requires greater training and accountability across all levels of the organisation.

Voluntary AI Safety Standard



Guardrail 8:

Be transparent with other organisations across the AI supply chain about data, models and systems to help them effectively address risks.

Governance and legal frameworks

Effective governance requires anticipation of AI's broader impacts, necessitating structures that can adapt to change. As AI applications permeate various sectors, organisations must ensure their governance frameworks can handle evolving legal and market dynamics. The role of voluntary guidance in shaping responses to AI challenges is another crucial consideration for proactive governance planning.

AI is currently regulated in a non-specific way in Australia by many existing laws - including the Privacy Act 1988 (Cth), Competition and Consumer Act 2010 (Cth), Copyright Act 1968 (Cth), Online Safety Act 2021 (Cth), Corporations Act 2001 (Cth), administrative laws, anti-discrimination laws, sector-specific laws and general laws like contract and tort laws. Establishing a clear understanding of these laws and how they apply to AI is critical to ensure compliance.

The Australian Government is also proposing to introduce AI specific regulation and various AI related reforms to existing laws.

Given the application of existing laws to AI, a solid foundation of compliance infrastructure within organisations is likely to be in place already. That should be reviewed for its application to AI. Then, AI specific issues and cross framework integration can be handled by an AI specific framework.

Supply chain risk assessment

The establishment of common industry standards and frameworks is essential for facilitating negotiations and ensuring equitable agreements between parties of varying sizes. Adopting global best practices to update existing frameworks with AI-specific elements is crucial. For example, traditional supplier risk management evaluates vendors based on financial stability and compliance with general business regulations. However, AI-specific supplier risk management expands these evaluations to address unique AI challenges such as ethical data usage, algorithmic transparency and bias mitigation. It involves auditing AI development processes, scrutinising the data sets used for training algorithms, and continuously monitoring AI outputs to ensure they adhere to ethical standards and regulatory requirements. This comprehensive approach ensures that third-party vendors meet *stringent standards* necessary for the safe and ethical deployment of AI technologies to align with both existing regulations and emerging standards specific to AI.

Complexities in AI procurement and contractual issues

The procurement of AI technologies presents some unique challenges that differ from those associated with more traditional rule based technologies. This requires rethinking some of the traditional approaches to procurement such as how to develop requirements specifications (favouring an outcome-based approach), iterative development, testing processes and addressing data provenance and management. Procurement contracts should be written to handle these new approaches also address any AI-specific risks - for example, the risk of copyright or privacy infringement associated with the use or creation of data using generative AI.



Expert tip:

Establish an understanding of existing law and how they apply to AI. Review existing compliance frameworks to make sure they work with AI.

Also, implement an AI policy and compliance framework that deals with AI-specific issues and integrates with existing frameworks.

Voluntary AI Safety Standard



Guardrail 7:

Establish processes for people impacted by AI systems to challenge use or outcomes.

Contracts for the deployment of AI should explicitly incorporate ethical considerations and values to proactively address risks such as bias, security and transparency. This ensures that *accountability measures* are embedded throughout the AI lifecycle, from development to deployment and operation, enhancing both compliance and ethical integrity in AI applications.

Addressing the challenge of non-specialist understanding

It is crucial that AI risks are *understandable* to a broad audience. Simplifying complex technical details without losing the necessary depth requires careful communication strategies that enhance transparency and facilitate broader engagement.

Balancing AI opportunities with risk mitigation

AI offers transformative opportunities for businesses and society, though it comes with inherent risks that necessitate strategic management. By embracing AI opportunities strategically and managing associated risks diligently, organisations can not only improve their operational efficiencies but also drive innovation responsibly.

Balancing these aspects requires continuous assessment, adaptive strategies and an inclusive approach to AI integration, ensuring AI deployments enhance capabilities while safeguarding against potential risks.

Strategic adoption and deployment

The potential of AI to catalyse significant business growth is increasingly acknowledged across industries. Adopting AI strategically, with a balanced focus on both its risks and opportunities, is critical in cultivating a culture that prioritises responsible exploration and innovation. Tools like Microsoft Copilot, for instance, are being integrated into business workflows to enhance productivity and decision-making processes. This showcases how AI can be seamlessly incorporated into daily operations to drive efficiency and competitive advantages, demonstrating the strategic integration that yields substantial benefits.

To ensure this integration is both effective and secure, organisations must implement a *proactive risk management strategy*. This strategy extends beyond addressing immediate risks, such as data breaches or ethical concerns, to include the anticipation of long-term challenges like the impact of AI on business processes and compliance requirements. Effective risk management requires continuous evaluation of AI systems for vulnerabilities, enabling organisations to adapt to new threats and maintain control over AI operations. By aligning risk management with strategic deployment, organisations can safely and effectively harness AI's capabilities, leading to *progressive advancements* rather than stagnation.



Expert tip:

Embed AI related legal, ethical and commercial considerations and practices by design into AI procurement and contracts.

Voluntary AI Safety Standard



Guardrail 10:

Engage your stakeholders and evaluate their needs and circumstances, with a focus on safety, diversity, inclusion and fairness.

Adopting a holistic approach to AI deployment involves forming *multi-functional teams* to comprehensively scrutinise AI applications.

Developing guidelines for responsible AI use, especially in customer-facing and internal business operations, is vital for compliance without compromising operational integrity.

The strategic appointment of people with diverse viewpoints, including sceptics, for AI initiatives can enrich governance processes, ensuring rigorous scrutiny and a balanced perspective on AI's potential and pitfalls.

By integrating strategic deployment, risk management and comprehensive governance, organisations can achieve a robust framework that not only supports innovative AI applications but also ensures they are ethically sound and compliant with regulatory standards. This integrated approach ensures that AI deployments enhance capabilities while safeguarding against potential risks, promoting a responsible and sustainable advancement in AI technology.

If you approach something with a 'what can we do, how do we do this responsibly' mindset, it changes the dynamic.

– Kenneth Weldin

Expanding AI opportunities across diverse sectors

AI is playing a transformative role across various sectors, including the not-for-profit sector, where its impact is particularly notable in enhancing operational efficiency even in resource-constrained environments. In these settings, AI excels in data analysis, streamlining governance processes and facilitating nuanced policy development. For instance, AI tools can automate the analysis of large volumes of donor data to identify trends and insights that drive more targeted and effective fundraising strategies. AI can be employed to monitor and report on governance practices, ensuring compliance and transparency to strengthen donor trust and engagement. These capabilities not only revolutionise how services are provided and how donor relationships are managed but also enhance overall organisational efficiency, making AI a valuable asset in extending the reach and impact of not-for-profits.



Expert tip:

Implement a continuous evaluation model to detect AI system vulnerabilities and threats to maintain effective control over AI operations.

3. AI as an enabler

AI's enabling power spans diverse applications and fields. Unlike traditional technologies, AI possesses the ability to learn, adapt, and perform tasks that typically require human intelligence, making it a versatile tool for organisations seeking to stay competitive in a rapidly evolving market.

AI fosters collaboration between humans and machines, *augmenting* human abilities and enabling new forms of interaction and problem-solving. This synergy can democratise access to information and resources, empowering organisations with limited capabilities to achieve their missions more effectively. In sectors such as healthcare, education and not-for-profit, AI can bridge gaps and provide tailored solutions that were previously unattainable. Laborious time-consuming and repetitive activities can now be replaced with further human creativity and innovation.

AI is a complementary tool to assist in human labour that will augment human activities to higher value-add activities.

Recognising AI as an enabler involves understanding its strategic importance, the necessity for robust implementation frameworks and the ethical considerations essential for responsible deployment. AI still requires *human supervision* and judgement. By leveraging AI's capabilities, organisations can enhance their operational efficiencies and drive meaningful societal impact. This ensures sustainable and inclusive growth in the digital age, fostering innovation while addressing global challenges.

Building the business case and value proposition for AI

Establishing a compelling business case for AI involves demonstrating its potential to deliver significant value across various organisational dimensions, including enhancing operational efficiencies, improving decision-making processes and fostering innovation.

Strategic importance and urgency

Business leaders are increasingly recognising the strategic importance of AI in maintaining a competitive edge and in fostering innovation. This strategic move is driven by an era of hyper-personalisation that aims to tailor products and services to better meet consumer needs, along with enhanced supply chain integrity and transparency to achieve ESG goals and meet stakeholder expectations.



Expert tip:

Assess the strategic importance and imperative of AI adoption across the business and the broader industry.

Voluntary AI Safety Standard



Guardrail 5:

Enable human control or intervention in an AI system to achieve meaningful human oversight across the life cycle.

AI's role goes beyond just automating tasks. It enhances the agility of an organisation, allowing it to respond more swiftly to market forces, including evolving consumer needs.

AI plays a crucial role in the development of both current employees and the future workforce by integrating advanced technologies into educational programs and workplace software applications. These initiatives not only prepare individuals for a digital future but also ensure that organisations are continually revitalised with fresh ideas and new approaches to problem-solving. To effectively harness AI's potential, however, organisations should adopt an incremental integration approach. This involves facilitating careful evaluation and governance, with a realistic understanding of AI's capabilities and limitations so its application is both ethical and effective.

Enhancing operational efficiency and security

AI's transformative capabilities are evident in sectors like healthcare. In healthcare, AI tools play a pivotal role by analysing vast amounts of patient data to forecast health outcomes and tailor treatment plans to individual needs. This not only boosts the efficiency of healthcare delivery but also elevates the quality of patient care.

In the realm of security, AI-enhanced surveillance systems are instrumental. These systems scrutinise behaviour patterns to detect anomalies and promptly alert human operators, thereby averting potential security threats before they escalate. Through these innovations, AI is not just streamlining operations but is also bolstering safety and security across diverse fields.

Accelerating startup development

Startups, especially in the technology sector, are leveraging AI to accelerate their development processes and enhance market analysis. By incorporating AI, these nimble businesses can automate significant portions of manual processes. This automation allows businesses to allocate more time and attention to other aspects, such as refining design and user experience, which can differentiate a product in competitive markets. The use of AI not only fast-tracks the product development cycle but also contributes to substantial reductions in labour costs. These efficiencies are crucial for startups, where speed and cost-effectiveness are often paramount to success. The *speed of market access and penetration* is the new competitive advantage.

Startups are demonstrating a strong willingness and a greater risk appetite when it comes to adopting AI. AI helps them get products to market faster – usually half the time and half the cost.

– Michelle Moffatt



Expert tip:

Leverage AI tools to enhance the monitoring and mitigating security of digital assets and online environments.

Facilitating sustainable practices and economic efficiency

In sectors such as advanced manufacturing, AI is becoming a key player in promoting sustainability and enhancing economic efficiency. AI enables companies to optimise their energy consumption and minimise waste through smarter management of resources. For example, AI can help with predictive maintenance, which not only prevents unexpected equipment failures but also extends the lifespan of the machinery, conserving resources and reducing waste. AI can help fine-tune supply chain operations, ensuring that materials and products are moved more efficiently, further cutting down on excess and inefficiency. Together, these applications of AI support a more sustainable manufacturing process, aligning operational goals with environmental stewardship.

Generative AI: lowering barriers and encouraging innovation

Generative AI represents a significant evolution towards more advanced AI systems. Unlike its predecessors, generative AI can create *new content*, from textual outputs to images and code, demonstrating a leap in AI's creative and functional capabilities. ChatGPT is a prominent example of generative AI, known for producing coherent and contextually relevant text in a conversational style. It is an example of the democratisation of AI.

This capability has broadened the scope of AI applications in everyday tasks and business operations.

In parallel, APIs provide accessible and streamlined access to powerful generative AI models, enabling developers and businesses to easily integrate these capabilities into their own applications. This empowers even small businesses and individual entrepreneurs, who can now deploy advanced AI technologies without the need for extensive resources or deep technical expertise. By making these tools widely accessible, generative AI enhances creative processes and efficiency, fostering innovation and providing a competitive edge across various industries.

Small business operators should still seek to understand how AI may affect their business decisions, legal compliance and operations.

Some particular considerations for creators and innovators include:

- **Ownership of content:** Copyright generally does not subsist in AI-generated works in Australia because copyright law requires there to be a human author. So, creators may not own the copyright in their content if it was generated by AI. Others may then use that content without infringing copyright. Similarly, patent law requires a human inventor, so an invention generated by AI may not be patentable.
- **Liability for output of an AI:** AI may produce output that is inaccurate, unreliable, misleading, biased, defamatory or otherwise inappropriate. The risk increases where AI is used to make decisions or there is no effective human review. The fact that AI produced the output is unlikely to be a legal defence.



Expert tip:

Collaborate with your organisation's team leaders to identify and recognise the benefits AI deployment across the supply chain, starting with pre-production services to post-production activities.

- **Use of third party data:** Organisations should ensure they have appropriate rights (from a copyright, confidentiality and privacy perspective) before providing data to an AI system.
- **Protecting data from training:** AI systems are often trained on a large quantity of data scraped from the internet. Organisations should consider if they are comfortable with their data being viewed by AI models and, if not, taking technical or legal steps to prevent that from occurring.

The barrier for adoption has really come down...the cost to get going with experiments and using the technologies is very low, the amount of skill required to engage with the technologies also just dropped away.

Chris Burling

Open-source AI: opportunities and challenges

Open-source AI has revolutionised the way organisations, from startups to multinational corporations, access and use advanced technology. However, while it provides transformative opportunities for technological advancement and collaborative innovation, effectively navigating its landscape demands a deeper understanding of both its benefits and inherent risks.

By making AI models and frameworks *publicly available*, it enables a global community of developers to innovate and accelerate technological advancements. For instance, open sourcing pre-trained AI models empowers a global community of developers to innovate and tailor these models to their specific needs through fine-tuning. This approach not only democratises access to cutting-edge technology but also accelerates the pace of innovation across and between various sectors, enabling customised solutions that address unique challenges and enhance competitive advantages.

However, the rapid development cycle of using open-source models, while advantageous, presents its own set of challenges when compared to developing and deploying proprietary systems.



Expert tip:

Prioritise the robustness of data privacy, cyber security and backup systems in the initial stages of AI technology adoption to safeguard against immediate and long-term operational risks.

Open-source AI allows for quicker adaptation and integration, offering a significant edge over slower, closed-source developments from scratch. Yet, this can sometimes result in less rigorous testing and variability in quality. Organisations must balance the speed of adoption with thorough validation processes to ensure reliability.

Startups face distinct challenges when integrating open-source AI, particularly due to their reliance on external AI services which may experience interruptions. To mitigate such vulnerabilities, it is advisable for startups to develop diversified technology strategies. These should



Guardrail 9:

Keep and maintain records to allow third parties to assess compliance with guardrails.

encompass a variety of open-source options, alongside alternative solutions and comprehensive [backup systems](#). Drawing on insights from the volatile crypto industry, it is essential for startups to prioritise robustness and data privacy from the initial stages of technology adoption. This approach not only safeguards against immediate operational risks but also establishes a foundation for long-term resilience and sustained growth.

Organisational readiness for AI

Getting an organisation ready for AI will help harness the full potential of AI while aligning its use with broader strategic objectives.

Technological infrastructure

Successful AI adoption requires a robust infrastructure tailored to the demands of AI technologies. Organisations must assess their [current systems](#) to identify gaps, focusing on enhancing data processing capabilities and security measures. Upgrading hardware and implementing advanced encryption and automated bias detection tools are essential for responsible AI practices. Alternatively, organisations can leverage [cloud-based services](#), which offer scalable and cost-effective solutions for AI deployment without the need for extensive on-site upgrades. This approach simplifies the integration of advanced AI technologies and ensures compliance with evolving data protection and privacy standards.

Workforce preparedness

Preparing the workforce for AI integration involves equipping employees with the [technical skills](#) necessary for working with AI technologies and a deep understanding of [responsible AI](#).

Continuous education and training programs are essential for developing AI literacy across the organisation.

These programs should focus on fostering a culture of responsible innovation, ensuring employees are proficient in using AI tools while understanding the ethical considerations and best practices for deploying AI responsibly.

Cultural readiness and leadership

Fostering a culture that promotes innovation, agility and responsible AI practices is essential for successful AI integration. Organisations that encourage collaboration, experimentation and continuous improvement are better equipped to implement AI effectively. [Strong leadership](#) plays a crucial role in this process by championing AI initiatives and establishing governance frameworks that cover risk management and ethical considerations. Embedding responsible AI principles into the culture ensures that ethical considerations are central to AI projects, enhancing both productivity and ethical standards. Leaders must ensure that these values are upheld throughout the organisation, guiding teams toward ethical and innovative AI solutions.



Expert tip:

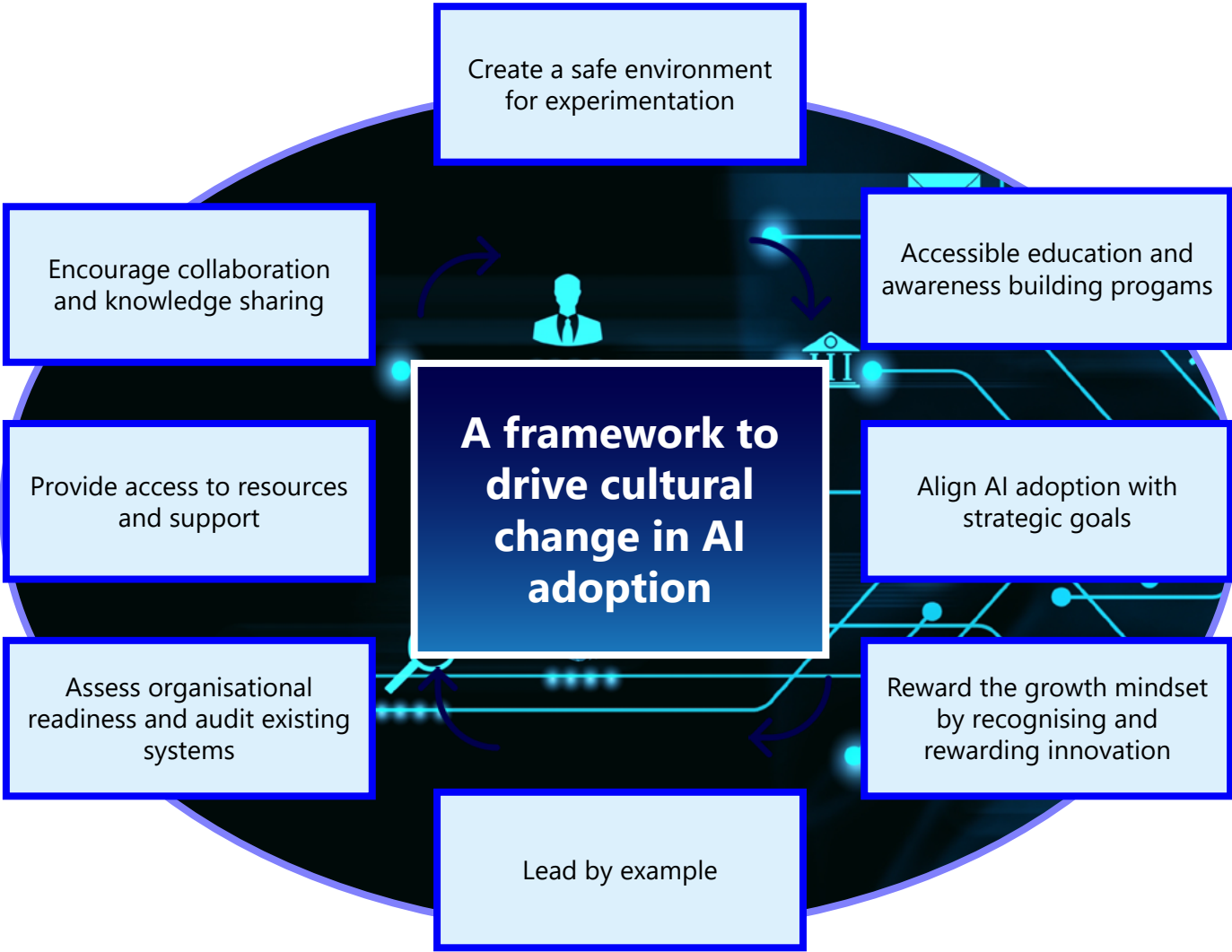
Align AI training with existing digital literacy training programs such as cyber security training to uplift capabilities across the organisation.



Expert tip:

Develop and apply governance frameworks that drive cultural change in AI adoption within teams and across the organisation.

Figure 3: Driving cultural change across the organisation



Best practices: taking a tailored approach to meet organisational needs

Successfully integrating AI into an organisation requires a comprehensive strategy encompassing proactive risk management, ethical governance, continuous learning and culture change. There is no one-size-fits-all approach. What may work well for one organisation may not work for others.

Leveraging [reusable patterns](#), such as those highlighted in the Voluntary AI Safety Standard, RAI patterns¹⁷ and process metrics¹⁸, can streamline the adoption process and ensure best practices are followed. The following best practice recommendations aim to guide organisations in effectively adopting and managing AI technologies while ensuring alignment with broader organisational goals and ethical standards.

A checklist for AI governance

- ❑ **Adopt a proactive approach:** Organisations should continuously learn and adapt to the evolving AI landscape. This involves fostering a culture of proactive risk management and ethical AI deployment. By staying ahead of technological advancements and potential risks, organisations can better navigate the complexities of AI integration.
- ❑ **Integrate AI into existing frameworks:** AI considerations should be integrated into existing organisational structures such as ESG frameworks. This ensures that AI deployment aligns with broader organisational and governance goals. Add AI specific frameworks on top to address AI specific issues and to ensure that there are no gaps. Standards such as ISO42001:2023 Artificial Intelligence management system and the Australian Government's Voluntary AI Safety Standard can help with this.
- ❑ **Enhance digital literacy and education:** Boards and executives must prioritise digital literacy and upskilling to effectively engage with AI-related issues. Continuous education and training programs are essential to equip employees with the necessary skills and understanding of AI technologies. These programs should focus on developing intuitive standards and clear guidance for both technical and non-technical users, fostering a knowledgeable and adaptable workforce.
- ❑ **Adopt the Voluntary AI Safety Standard:** Adopt comprehensive governance frameworks that address both internal and third-party AI use. The Voluntary AI Safety Standard supports safe and responsible AI deployment across all levels of the organisation, including suppliers and partners. Integrate AI considerations into existing governance frameworks and policies, such as developing a hardened sandbox environment for testing AI systems and ensuring comprehensive supplier risk management.
- ❑ **Foster collaboration and adhere to standards:** Global collaboration and adherence to international standards are crucial for effective AI governance. Aligning with frameworks such as the EU AI Act and other international guidelines ensures responsible AI deployment and promotes consistency across different jurisdictions. Collaboration also involves sharing best practices and leveraging reusable patterns like RAI to streamline AI adoption.

¹⁷ CSIRO, 2024, [Responsible AI Pattern Catalogue](#) [web page].

¹⁸ Xia B, Lu Q, Zhu L, Lee SU, Liu Y and Xing Z, 2023, 'Towards a Responsible AI Metrics Catalogue: A Collection of Metrics for AI Accountability', arXiv preprint arXiv:2311.13158.

Conclusion

The three sections of this white paper highlight the multifaceted nature of AI, emphasising its potential to drive innovation, enhance operational efficiency, and address complex societal and environmental challenges. However, the adoption of AI also brings significant risks and ethical considerations that organisations must appropriately identify and navigate.

The roundtable discussions reinforced that responsible AI adoption requires a holistic approach, integrating ethical considerations, robust governance, proactive risk management, strategic innovation and culture change. By adopting best practices and aligning with international standards, organisations can harness AI's transformative benefits while ensuring compliance with ethical and regulatory requirements. This comprehensive approach will enable organisations to leverage AI for strategic success, driving innovation, operational excellence and positive societal impact.



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